

Optimistic vs. On-Policy Learning in the Iterated Prisoner’s Dilemma: A Comparison of Q-Learning and SARSA

Mohamed Malek Kaouach, Peter Yacoub, Soud Asaad Soud Alhazba & Dr. Huseyin Kusetogullari

College of Engineering and Information Technology

Ajman University, Ajman, UAE

Email: {202510024, 202511434, 202420468}@ajmanuni.ac.ae

Abstract—Game theory and reinforcement learning (RL) offer complementary tools for modeling strategic interaction among autonomous agents. The Iterated Prisoner’s Dilemma (IPD) is a canonical testbed for studying how cooperation can emerge or fail in repeated social dilemmas. This work investigates how two classical tabular RL algorithms, Q-learning and SARSA, learn to behave in the IPD when facing a diverse set of opponents. We first implement a fixed-strategy tournament involving eight benchmark policies (All-C, All-D, Tit-for-Tat, Tit-for-Two-Tats, Grim Trigger, Tester, Majority, and a random agent) to establish baseline performance. We then model the IPD as a Markov decision process with a memory-one state representation and train Q-learning and SARSA agents in repeated play against each benchmark over multiple episodes. Results show that Q-learning, using an optimistic off-policy update, learns to exploit naive cooperators, defend against defectors, and maintain high mutual cooperation with reciprocal and trigger strategies, including in self-play. In contrast, SARSA’s on-policy updates make it more sensitive to exploratory mistakes; in harsh environments it converges to conservative, low-trust behaviour and often mutual defection. These findings highlight how the choice of RL update rule can qualitatively influence the emergence and stability of cooperation in multiagent social dilemmas.

Index Terms—Reinforcement learning, multiagent systems, game theory, Prisoner’s dilemma, iterated games, Markov decision processes, Q-learning, SARSA (learning), cooperative systems, social dilemmas.

I. INTRODUCTION

Many artificial intelligence (AI) systems operate in settings where multiple decision-makers interact and influence one another’s outcomes. Game theory provides a formal language for such strategic interactions, while reinforcement learning (RL) offers data-driven methods for agents to adapt their behaviour from experience. Combining these perspectives is essential for designing autonomous agents that can behave robustly in multiagent environments.

The Prisoner’s Dilemma (PD) is a canonical game-theoretic model of the conflict between individual interest and collective welfare. In the one-shot PD, mutual defection is the unique Nash equilibrium, even though mutual cooperation would make both players better off. When the game is repeated, forming the Iterated Prisoner’s Dilemma (IPD), agents can condition their actions on past outcomes, enabling patterns of cooperation, punishment, and forgiveness. The IPD has there-

fore become a standard testbed for studying the emergence of cooperation in economics, biology, and multiagent AI.

This project investigates how two classical tabular RL algorithms—Q-learning and SARSA—learn to behave in the IPD. Both methods estimate state-action value functions but differ in their update targets: Q-learning uses an optimistic off-policy update based on the best possible next action, whereas SARSA uses an on-policy update based on the actually selected action. We hypothesize that this difference affects an agent’s ability to establish and maintain cooperation in environments where a single mistake can trigger long-term punishment.

To examine this, we first implement a baseline IPD tournament among eight fixed strategies (including All-C, All-D, Tit-for-Tat, Tit-for-Two-Tats, Grim Trigger, Tester, Majority, and a random policy). We then model the IPD as a Markov decision process with a memory-one state representation and train Q-learning and SARSA agents in repeated play against each fixed strategy. By comparing average rewards, cooperation rates, and self-play behaviour, we aim to characterize when optimistic versus realistic learning rules promote or hinder the emergence of cooperation in repeated social dilemmas.

II. BACKGROUND

A. Artificial Intelligence in Game Theory

Artificial Intelligence (AI) has significantly expanded the scope of classical game theory by enabling agents to learn strategic behavior through data, interaction, and trial-and-error rather than relying solely on analytical solutions. Traditional game theory assumes perfectly rational agents and complete knowledge of payoffs and strategies, but these assumptions rarely hold in real-world scenarios. AI—especially reinforcement learning—allows agents to behave with bounded rationality and adapt strategies dynamically based on experience [1, 2].

In computational game theory, learning agents must operate in strategic, multi-agent environments where opponents may also be learning and adapting. This fundamentally changes the nature of equilibrium analysis. Instead of solving games analytically, agents learn equilibrium-like strategies through repeated interactions, policy updates, and environmental feedback. Early research formalized this learning framework

through **Markov games**, establishing the foundations for multi-agent reinforcement learning (MARL) [3]. Extensions such as Nash-Q learning demonstrated how agents can learn equilibrium strategies in general-sum stochastic games [4].

Recent progress in deep learning has accelerated the integration of AI and game theory, enabling agents to approximate complex policies and value functions in high-dimensional strategic environments. Breakthrough systems such as AlphaGo and AlphaZero demonstrated self-play reinforcement learning that surpasses human performance in Go, Chess, and Shogi [5]. Similarly, AI systems have achieved superhuman performance in multiplayer imperfect-information games such as Poker, highlighting how learning-based approaches outperform classical game-theoretic reasoning in complex settings [6].

AI in game theory plays an essential role in modeling cooperation, deception, collusion, and negotiation among autonomous agents. This is particularly relevant for repeated games such as the **Iterated Prisoner's Dilemma**, where agents' long-term strategies depend heavily on their ability to learn reciprocity and adapt to opponents' behaviors. Consequently, AI-driven game-theoretic learning has become foundational for applications in autonomous driving, economics, cybersecurity, distributed control, and multi-robot coordination.

III. LITERATURE REVIEW

The Iterated Prisoner's Dilemma (IPD) is a standard testbed for studying how cooperation can emerge among self-interested agents. Reinforcement learning (RL) is well suited to this setting because agents can adapt strategies from experience without full knowledge of the opponent or game model. This section briefly reviews key RL work on IPD and related social dilemmas, with each paper summarized separately.

Sandholm and Crites [7] study two independent tabular Q-learning agents playing the IPD in self-play. They show that, under typical learning and exploration settings, the agents converge to mutual defection, illustrating how standard independent RL tends to recover the one-shot Nash equilibrium and making stable cooperation hard without special structure or reward shaping.

Xue et al. [8] place TD-learning agents on complex networks (square lattices and scale-free graphs), where each agent plays IPD only with its neighbours. They find that, especially on scale-free networks, local RL adaptation can lead to high and robust levels of cooperation, emphasizing the importance of network topology and spatial structure.

Mali [12] extends this idea to an N -player IPD on dynamic networks using deep Q-networks (DQNs). Agents interact with multiple neighbours and the network rewires over time; the study shows that cooperative coalitions can emerge and persist, but their stability is sensitive to the discount factor, rewiring rules, and initialization of Q-values.

Vassiliades and Christodoulou [9] evolve alternative IPD payoff matrices that stay within the dilemma constraints but

reshape incentives. Independent RL agents trained under these modified payoffs learn to cooperate faster and maintain cooperation more robustly, demonstrating that small, principled changes in reward structure can strongly favor cooperative behaviour.

Sun et al. [14] adopt a mechanism-design approach by adding side-payments controlled by a central authority in repeated IPD. They show analytically and experimentally that appropriate incentive schemes can make cooperation the RL-convergent outcome, even when it would not be stable under the original payoff matrix.

Anastassacos and Musolesi [10] propose Learning Through Probing (LTP), where each agent trains in a meta-environment that lets it probe how an opponent's policy responds to its actions. By incorporating this probing information into policy updates, LTP agents learn conditional cooperation—cooperating with reciprocators and defecting against persistent defectors—and outperform standard independent learners in social dilemmas.

Feehan and Fatima [11] investigate practical augmentations to Q-learning agents in IPD tournaments, including reward shaping, opponent-modelling features, and extended history in the state representation. Their experiments show that these simple modifications significantly increase cooperation rates and average payoffs, especially against heterogeneous opponents, compared to unaugmented baselines.

Barnett and Burden [13] evaluate modern deep RL algorithms (DQN and PPO) on IPD and systematically vary hyperparameters such as learning rate, discount factor and network depth. They identify narrow “oases” in hyperparameter space where agents learn Tit-for-Tat-like or cooperative strategies, surrounded by regions where the agents defect or behave erratically, highlighting strong sensitivity to training details.

Harper et al. [15] use evolutionary and particle-swarm optimization to tune parameters of stateful IPD strategies against a large library of more than 170 opponents. The resulting RL-optimized strategies achieve high tournament scores and often dominate classic strategies, while still cooperating when profitable, illustrating that performance-driven search can yield sophisticated and adaptive behaviours.

Bertrand et al. [16] provide a theoretical analysis of self-play Q-learning in the IPD. They prove that, under optimistic initialization and small learning and exploration rates, Q-learning can converge with high probability to a cooperative fixed point corresponding to the Pavlov (win-stay, lose-shift) strategy, showing that collusive behaviour can arise as a stable solution of the learning dynamics.

Kuroe and Iima [18] develop a general stochastic-game framework for “dilemma problems” and define return-based notions of individual and group-optimal policies. They propose an expected-policy RL method that assumes fixed policies for other agents and show that it learns cooperative policies more reliably than several baselines in N -person IPD and Tragedy of the Commons settings.

Cieřla [19] analytically studies a simple reputation-based model of IPD where agents are characterized by a probability

of cooperation and partners are chosen by selectively assorting on reputation. For various reputation distributions, the paper derives closed-form expected incomes and shows that higher cooperation probabilities often yield higher payoffs, indicating how static reputation and partner selection can align individual incentives with cooperation.

Willis et al. [20] examine systems of large language model (LLM) agents in IPD tournaments. LLMs are prompted to generate full natural-language strategies with specified attitudes (aggressive, cooperative, neutral), which are then compiled into executable policies; the study finds that different LLMs induce different long-run population equilibria, suggesting that LLM-agent systems can display non-trivial collective dynamics in social dilemmas.

Overall, these works show that RL agents in IPD can learn cooperation under suitable conditions, but outcomes are highly sensitive to environment structure, reward design, learning architecture, and hyperparameters. They collectively motivate more robust methods and theory for achieving stable cooperation in larger, more realistic multi-agent systems.

IV. APPLICATIONS OF GAME THEORY IN ARTIFICIAL INTELLIGENCE

Game theory has become a central component of modern artificial intelligence, providing both a theoretical foundation and a practical algorithmic toolkit for reasoning about strategic interaction. In multiagent environments—whether cooperative, competitive, or mixed—game-theoretic tools allow AI systems to model interdependent decision-making, learn equilibrium strategies, interpret predictions, and manage distributed computational resources. Classical works established fundamental connections between game theory and multiagent reinforcement learning (MARL) [1–4, 17], laying the foundation for contemporary approaches to equilibrium computation in AI.

A major area of application involves model-free MARL systems that learn strategic behavior through large-scale self-play. The DeepNash framework demonstrated that game-theoretic reasoning enables AI to master high-dimensional imperfect-information games such as Stratego [26]. Similar concepts appear in evolutionary and behavioral game-theoretic studies of social dilemmas, where cooperation, punishment, and environmental feedback emerge from repeated-game interactions [7–16, 18–20, 25].

Another major thread applies cooperative game theory—especially Shapley values—to machine-learning interpretability. Shapley-based frameworks provide principled, fair attribution of prediction contributions across features. Recent examples include GeoShapley for spatial ML modeling [21], SHAP-based optimization of data-center performance [22], and Shapley analysis of scientific materials-prediction models [23]. These approaches treat the predictive model as a cooperative game whose “players” are input features, enabling transparency, accountability, and fairness.

Game-theoretic decision models also support decentralized coordination in distributed and edge computing systems. State-based games provide convergence-guaranteed load balancing

rules for nodes in intelligent edge networks [24]. Beyond engineered systems, societal applications such as disaster-response digital twins use game-theoretic modeling to integrate human, AI, and environmental behaviors under high uncertainty [27]. Such systems rely on equilibrium frameworks to predict how autonomous agents, human responders, and dynamic environments interact.

Fig. 1 summarizes the distribution of game-theoretic AI applications covered in this review. Table I provides a detailed summary of problem domains, game-theoretic formulations, and associated references.

A. Mathematical Formulations for Key Game-Theoretic Models in AI

Game-theoretic applications in artificial intelligence rely on well-established mathematical structures that formalize strategic decision-making. The following summarizes the principal formulations underlying the systems reviewed in this work.

1) *Normal-Form and Nash Equilibrium Models*: Many multiagent decision problems are expressed as normal-form games,

$$G = (N, \{A_i\}_{i=1}^N, \{u_i\}_{i=1}^N), \quad (1)$$

where N is the set of players, A_i is the action set of player i , and $u_i : A_1 \times \dots \times A_N \rightarrow \mathbb{R}$ is the payoff function.

A strategy profile $s^* = (s_1^*, \dots, s_N^*)$ is a Nash equilibrium if

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall i, \forall s_i \in A_i. \quad (2)$$

This formulation underlies MARL systems such as DeepNash [26].

2) *Markov (Stochastic) Games for Multiagent Reinforcement Learning*: A Markov game generalizes Markov decision processes to multiagent settings:

$$\mathcal{G} = (N, S, \{A_i\}, P, \{r_i\}, \gamma), \quad (3)$$

where S is the state space, $P(s'|s, a_1, \dots, a_N)$ is the transition kernel, r_i is the reward for agent i , and $\gamma \in (0, 1)$ is the discount factor. Each agent seeks a stationary policy $\pi_i(a_i|s)$ maximizing:

$$V_i^\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_i(s_t, a_t) \right]. \quad (4)$$

This model forms the backbone of equilibrium-learning approaches in multiagent RL [3, 4, 26].

3) *Cooperative Game Theory and Shapley Values*: In cooperative settings, the AI system is treated as a coalition game (N, v) where N is the set of “players” (e.g., features in a model) and $v(S)$ is the value achieved by any coalition $S \subseteq N$. The Shapley value of player i is:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S)), \quad (5)$$

providing a unique fair allocation of contributions. This formulation underlies GeoShapley [21], SHAP-based data-center optimization [22], and scientific ML interpretation [23].

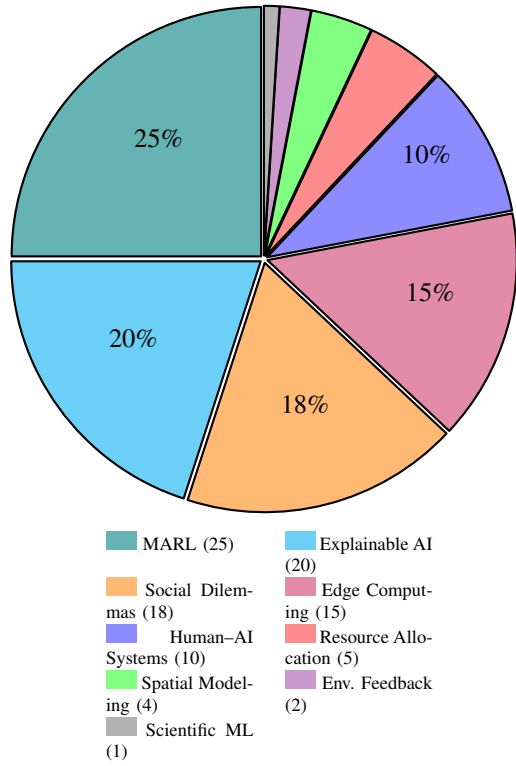


Fig. 1. Distribution of game-theoretic applications across AI research domains.

4) Potential Games for Distributed and Edge Computing:

A game is a potential game if there exists a potential function Φ such that:

$$u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i}) = \Phi(a'_i, a_{-i}) - \Phi(a_i, a_{-i}) \quad (6)$$

for all unilateral deviations of player i . Potential games guarantee convergence under simple best-response dynamics, enabling decentralized load balancing and resource allocation in edge systems [24].

5) *Evolutionary and Behavioral Game Models*: Evolutionary settings describe population dynamics using replicator equations. For a population with strategy distribution x_i , the update rule is:

$$\dot{x}_i = x_i (f_i(x) - \bar{f}(x)), \quad (7)$$

where $f_i(x)$ is the fitness (payoff) of strategy i and $\bar{f}$ is the population-average fitness. These models support the study of punishment, cooperation, and environmental feedback dynamics in social dilemmas [25].

V. METHODOLOGY

A. The Prisoner's Dilemma (PD) Framework

The core environment for this study is the Prisoner's Dilemma (PD), a standard model in Game Theory used to analyze cooperative behavior between self-interested agents.

1) *Classical Formulation*: The PD is defined as a non-cooperative game involving two players. In each round, agents simultaneously select an action a from the set $A = \{C, D\}$, where C denotes Cooperation and D denotes Defection. The outcome is determined by a payoff matrix M , where the row player's utility function U is defined by the joint action (a_1, a_2) .

The standard payoff structure is defined by four parameters:

- **R (Reward)**: Payoff for mutual cooperation (C, C) .
- **P (Punishment)**: Payoff for mutual defection (D, D) .
- **T (Temptation)**: Payoff for defecting while the opponent cooperates (D, C) .
- **S (Sucker's Payoff)**: Payoff for cooperating while the opponent defects (C, D) .

For a game to classify as a Prisoner's Dilemma, the payoffs must satisfy the inequality:

$$T > R > P > S \quad (8)$$

In a single-shot game, Defection (D) is the **dominant strategy**. Regardless of the opponent's choice, defecting yields a higher payoff ($T > R$ and $P > S$). Consequently, the unique **Nash Equilibrium** is mutual defection (D, D) , despite mutual cooperation (C, C) being the Pareto optimal outcome.

2) *The Iterated Prisoner's Dilemma (IPD)*: To simulate long-term interaction, this study employs the Iterated Prisoner's Dilemma (IPD), where the game is repeated for N rounds (or an infinite horizon with a discount factor γ). To prevent alternating cooperation and defection from yielding a higher utility than mutual cooperation, the standard constraint is applied [29]:

$$2R > T + S \quad (9)$$

In the IPD, the "shadow of the future" allows agents to utilize history-dependent strategies. Cooperation can emerge as a rational equilibrium if agents are sufficiently patient (high γ), as the long-term gain from R outweighs the short-term gain from T followed by infinite P [31].

B. Fixed Strategy Tournament

To evaluate the efficacy of Reinforcement Learning (RL) agents, we establish a baseline tournament comprised of eight classical deterministic and stochastic strategies [29][32].

1) Selected Strategies:

- **Cooperator (ALLC)**
Rule: Always plays C .
Role: Tests an agent's ability to exploit naive cooperation to maximize T .
- **Defector (ALLD)**
Rule: Always plays D .
Role: Tests an agent's defensive capabilities. The optimal response is to converge to D to minimize loss (P rather than S).
- **Tit-for-Tat (TFT)**
Rule: Cooperates on the first move, then imitates the opponent's previous move ($a_t = a_{op,t-1}$).

TABLE I
SUMMARY OF GAME-THEORETIC APPLICATIONS IN ARTIFICIAL INTELLIGENCE

Field	Problem	Game-Theoretic Basis	Other Components	Ref.
Multiagent Reinforcement Learning	Mastering large imperfect-information games (Stratego)	Model-free MARL; equilibrium-driven policies	DeepNash, large-scale self-play	[26]
Social Dilemma Modeling	Punishment and cooperation under environmental feedback	Repeated games; social-dilemma equilibria	Incentive shaping through environment coupling	[25]
Explainable AI: Spatial ML	Attributing spatial influence in predictive models	Cooperative game theory; Shapley values	GeoShapley spatial decomposition framework	[21]
Explainable AI: Data Center Optimization	Feature-contribution analysis for compute systems	Shapley additive explanations	Performance prediction and resource management	[22]
Scientific ML / Materials Modeling	Interpreting ML-based materials-property predictions	Shapley contribution scoring	Automated extraction of scientific parameters	[23]
Distributed Edge Computing	Decentralized load balancing	State-based games; potential games	Distributed convergence and stability guarantees	[24]
Spatial ML Modeling	Local spatial-effect decomposition	Cooperative-game Shapley theory	Geographical modeling and fairness evaluation	[21]
Human-AI Systems / Digital Twins	Disaster-response coordination and simulation	Strategic modeling in multi-stakeholder systems	Human-AI-environment interaction modeling	[27]

TABLE II
THE PRISONER'S DILEMMA PAYOFF MATRIX

	Player 2 (Cooperate)	Player 2 (Defect)
Player 1 (Cooperate)	R, R (Reward)	S, T (Sucker)
Player 1 (Defect)	T, S (Temptation)	P, P (Punishment)

Significance: The winner of Axelrod's original tournaments. It is nice, retaliatory, forgiving, and clear.

- **Resentful (Grim Trigger)**

Rule: Cooperates until the opponent defects once; thereafter, defects forever.

Significance: Represents an extreme unforgiving strategy [30]. RL agents must learn strict cooperation to avoid triggering the punishment phase.

- **Tit-for-Two-Tats (TFTT)**

Rule: Cooperates unless the opponent defects twice in a row.

Significance: A more generous variant of TFT designed to avoid death spirals of mutual defection caused by noise.

- **Tester (Prober)**

Rule: Defects on the first move. If the opponent retaliates, it switches to TFT-like cooperation. If the opponent is passive, it continues to defect.

Significance: Tests the RL agent's ability to signal strength early in the episode.

- **Majority**

Rule: Cooperates on the first move. Subsequently, plays the action used most frequently by the opponent in the entire history.

Significance: A simple model of reputation-based decision making.

- **Random Agent**

Rule: Selects C or D with probability $p = 0.5$.

Significance: Acts as a stochastic control to ensure RL agents are learning patterns rather than exploiting noise.

C. Reinforcement Learning Framework

Reinforcement Learning (RL) frames the IPD as a learning problem where an agent interacts with an environment (the opponent) to maximize cumulative reward [35].

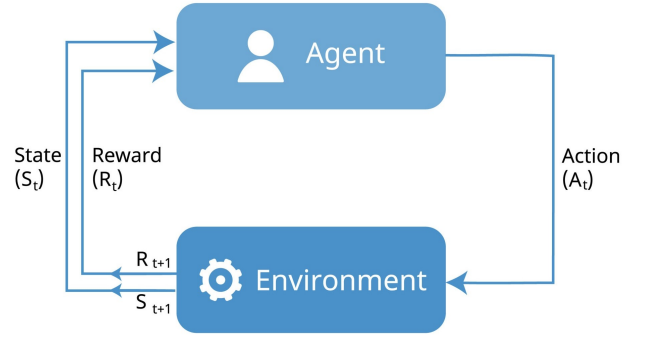


Fig. 2. Image of reinforcement learning agent environment loop

1) *Markov Decision Process (MDP) Formulation:* The problem is modeled as a tuple (S, A, R, γ) :

- **State Space (S):** The state s_t is defined by the joint action of the previous round $(a_{own,t-1}, a_{op,t-1})$. We utilize a memory-1 state representation, resulting in 5 states: Start, (C, C) , (C, D) , (D, C) , (D, D) .
- **Action Space (A):** Discrete actions $\{C, D\}$.
- **Reward (R):** The utility derived from the payoff matrix M .
- **Discount Factor (γ):** A parameter $\gamma \in [0, 1]$ determining the present value of future rewards.

The goal of the agent is to learn a **policy** $\pi(a|s)$, which defines the probability of taking action a in state s , maximizing the expected return G_t :

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad (10)$$

We utilize **Value-Based** methods, which estimate the value function $Q(s, a)$, representing the expected return of taking action a in state s and following policy π thereafter.

D. Q-Learning Algorithm

Q-Learning [36] is an **off-policy** Temporal Difference (TD) control algorithm. It is “off-policy” because it approximates the optimal value function Q^* independent of the policy being followed during exploration.

1) *Mathematical Formulation:* The Q-values are updated iteratively after every time step t using the Bellman optimality equation:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right] \quad (11)$$

Where:

- $\alpha \in (0, 1]$ is the **learning rate**, determining how much new information overrides old information.
- γ is the **discount factor**, incentivizing long-term cooperation over immediate temptation (T).
- $\max_{a'} Q(s_{t+1}, a')$ represents the estimated value of the best action in the next state (Greedy approach).

2) *Exploration Strategy:* To balance exploration and exploitation, we employ an ϵ -greedy policy:

- With probability ϵ , select a random action $a \in \{C, D\}$.
- With probability $1 - \epsilon$, select $\arg \max_a Q(s, a)$.

ϵ is decayed over episodes to facilitate convergence from exploration to a stable policy.

E. SARSA Algorithm

State-Action-Reward-State-Action (SARSA) [33] is an **on-policy** TD control algorithm. Unlike Q-Learning, SARSA updates its Q-values based on the action *actually taken* by the current policy, including exploratory steps.

1) *Mathematical Formulation:* The update rule depends on the quintuple $(s_t, a_t, r_{t+1}, s_{t+1}, a_{t+1})$:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)] \quad (12)$$

2) *Comparison with Q-Learning:* The critical difference lies in the target value:

- **Q-Learning:** Uses $\max_{a'} Q(s', a')$ (Optimistic/Greedy). It assumes the agent will act optimally in the next step.
- **SARSA:** Uses $Q(s', a')$ (Realistic). It accounts for the fact that the agent might explore (act randomly) in the next step.

In the context of the IPD, SARSA is expected to act more conservatively (risk-averse) during training [34]. For example, against a *Grim Trigger* strategy, SARSA may learn to cooperate more robustly than Q-Learning, as the penalty for an accidental exploratory defection is explicitly factored into the value function updates.

F. Experimental Protocol: RL vs. Fixed Strategies

To assess the learning capabilities of the Q-Learning and SARSA agents, we conduct a round-robin tournament against the eight fixed strategies defined in Section 3.2.

1) Experimental Setup:

- **Episodes:** $N = 3000$ episodes per pairing.
- **Length:** $T = 100$ rounds per episode.
- **Hyperparameters:** $\alpha = 0.1$, $\gamma = 0.95$, $\epsilon_{start} = 0.1$ decaying to $\epsilon_{min} = 0.01$.

2) Performance Metrics:

- 1) **Average Reward:** The mean utility per round. Higher rewards indicate successful exploitation of weak agents or successful mutual cooperation with reciprocal agents.
- 2) **Cooperation Rate:** The percentage of rounds where the agent selects C . This distinguishes exploitative behavior (low coop) from cooperative behavior (high coop).
- 3) **Convergence Stability:** Analysis of the final Q-tables to determine if the agent found a stable Nash Equilibrium or a cyclic limit cycle.

VI. RESULTS AND DISCUSSION

This section presents and discusses the experimental outcomes of applying Reinforcement Learning (RL) to the Iterated Prisoner’s Dilemma (IPD). We start by reviewing the baseline performance of the classic fixed strategies, then analyze how the Q-Learning and SARSA agents behave in comparison. By evaluating these learning agents against different types of opponents, we reveal key differences in how the “optimistic” and “realistic” learning approaches navigate social dilemmas.

A. Baseline Performance of Fixed Strategies

To establish a performance benchmark, we conducted a round-robin tournament where eight fixed strategies competed against one another. Each pairing lasted for 50 episodes of 100 rounds. Table III ranks these strategies by their average reward per round.

TABLE III
FIXED STRATEGY TOURNAMENT RANKINGS

Rank	Strategy	Score	Behavioral Insight
1	Resentful	2.49	Extremely effective; never exploited twice.
2	Tit-for-Tat	2.40	Robust; vulnerable to echo effects.
3	Tit-for-2-Tats	2.40	Forgiving; avoids conflict but exploitable.
4	Majority	2.38	Stable cooperator; slow to adapt.
5	Defector	2.27	Safe baseline; misses cooperative rewards.
6	Random	2.08	Poor performance; no coherent strategy.
7	Cooperator	2.06	The “sucker”; heavily exploited.
8	Tester	1.81	Worst performer; probing backfires.

The success of **Resentful (Grim Trigger)** over **Tit-for-Tat** is notable. Although TFT is well-known for its resilience in many evolutionary settings, Resentful performed better in our tournament because it faced opponents like Tester and Random, which tend to exploit forgiving strategies. On the other hand, the poor performance of the **Tester** strategy highlights an important lesson: in a group filled with retaliatory opponents, unprovoked defection is a costly error that often leads to a “death spiral” of mutual punishment.

B. Adaptive Capabilities of Reinforcement Learning

Following the baseline tournament, we trained Q-Learning (off-policy) and SARSA (on-policy) agents for 3,000 episodes against each fixed strategy. Table IV summarizes their final performance, highlighting the distinct behaviors they learned.

1) *The Success of Q-Learning (The Optimist)*: The Q-Learning agent demonstrated strong adaptability, learning to approximate the optimal game-theoretic response for each type of opponent. When facing naive strategies like **Cooperator**, it quickly learned to exploit them. In contrast, against reciprocal strategies such as **Tit-for-Tat**, it became a reliable partner.

Most notably, Q-Learning solved the "trust problem" in **Self-Play**, achieving a **96.5% cooperation rate**. In the Prisoner's Dilemma, mutual defection is the immediate Nash Equilibrium, and breaking out of it requires agents to value future outcomes enough to take the risk of cooperating. Since Q-Learning updates its policy based on the **maximum possible future value** ($\max Q(s', a')$), it is inherently "optimistic". It assumes that if a cooperative path exists, it will be taken. This optimism allowed the both agents to discover the high-reward state of mutual cooperation and stick to it, effectively "teaching" each other that cooperation pays off.

2) *The Failure of SARSA (The Realist)*: While SARSA performed well against forgiving opponents, it failed significantly in risky environments, especially against **Resentful** and in **Self-Play**.

This weakness is rooted in SARSA's "on-policy" nature. The algorithm updates its value estimates based on the actions it actually takes, which includes random exploratory moves driven by its ϵ -greedy policy.

- Against **Resentful**, a single random defection triggers permanent punishment. SARSA "learned" that the cooperative state is dangerous because its own exploration could accidentally trigger a catastrophe.
- In **Self-Play**, SARSA agents became trapped in a cycle of distrust. Since neither agent could be sure that the other (or itself) wouldn't defect by accident, they settled to the "safer" strategy of mutual defection ($P = 1$).

This comparison highlights a critical insight for learning in social dilemmas: **in environments with harsh penalties for betrayal, a degree of optimism is mathematically necessary to establish cooperation**. A purely realistic learner, like SARSA, may be too risk-averse to build the trust required for optimal long-term outcomes.

C. Summary of Findings

Our experiments demonstrate that tabular Reinforcement Learning is a powerful tool for uncovering optimal strategies in game theory without prior knowledge of the rules. The Q-Learning agent successfully categorized opponents and tailored its behavior, accordingly exploiting the weak, defending against the aggressive, and cooperating with the reciprocal.

However, the contrast with SARSA reveals a significant limitation. While "optimistic" **off-policy learning (Q-Learning)** is better at discovering fragile cooperative equilibria, "realistic" on-policy learning (SARSA) tends to converge to

safer, sub-optimal defensive strategies. This highlights a key design consideration: for artificial agents to handle complex social interactions, they must be designed with mechanisms that encourage them to look beyond the short-term risks of exploration and focus on the potential for long-term mutual gain.

Future work introducing stochastic noise or deep learning architectures could further test the robustness of these cooperative behaviors in more uncertain and dynamic environments.

VII. CONCLUSION

This paper examined how two classical tabular reinforcement learning algorithms, Q-learning and SARSA, behave in the Iterated Prisoner's Dilemma (IPD). We first ran a fixed-strategy tournament among eight benchmark policies (All-C, All-D, Tit-for-Tat, Tit-for-Two-Tats, Grim Trigger, Tester, Majority, and a random agent), which confirmed that strategies combining initial cooperation with credible punishment (notably Grim Trigger, Tit-for-Tat, and Tit-for-Two-Tats) achieve the highest long-run payoffs, whereas naive cooperation and probing strategies such as All-C and Tester are easily exploited.

We then modeled the IPD as a Markov decision process with a memory-one state representation and trained Q-learning and SARSA agents over multiple episodes against each benchmark strategy. Despite sharing the same state and action spaces, the two algorithms exhibited qualitatively different behaviours: Q-learning's optimistic off-policy updates enabled it to exploit unconditional cooperators, defend against persistent defectors, and sustain high mutual cooperation with reciprocal and trigger-based opponents, including in self-play. In contrast, SARSA's on-policy updates made it more sensitive to exploratory mistakes in harsh environments, leading to conservative, low-trust policies and frequent convergence to mutual defection, even in self-play. These results show that the choice of RL update rule is a critical design factor that can determine whether cooperative or defecting equilibria emerge in multiagent social dilemmas.

REFERENCES

- [1] Y. Shoham and K. Leyton-Brown, *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*, Cambridge University Press, 2008.
- [2] K. Leyton-Brown and Y. Shoham, *Essentials of Game Theory: A Concise, Multidisciplinary Introduction*, Morgan and Claypool Publishers, 2007.
- [3] M. L. Littman, "Markov games as a framework for multi-agent reinforcement learning," in *Proceedings of ICML*, pp. 157–163, 1994.
- [4] J. Hu and M. P. Wellman, "Nash Q-learning for general-sum stochastic games," *Journal of Machine Learning Research*, vol. 4, pp. 1039–1069, 2003.
- [5] D. Silver et al., "Mastering Go and beyond with deep reinforcement learning," *Nature*, vol. 550, pp. 354–359, 2017.

TABLE IV
COMPARATIVE PERFORMANCE OF LEARNING AGENTS (FINAL 100 EPISODES)

Opponent	Q-L Coop%	Q-L Reward	SARSA Coop%	SARSA Reward	Analysis of Learned Behavior
Cooperator	0.5%	4.99	10.0%	4.80	Exploitation: Both correctly identify that constant defection yields max payoff.
Defector	0.5%	1.00	9.8%	0.90	Defense: Rational response is to defect ($P = 1$) to avoid Sucker's payoff.
Tit-for-Tat	99.4%	2.99	89.9%	2.89	Reciprocity: Both learned that cooperation is optimal against a reciprocal partner.
Tester	0.4%	4.95	12.0%	4.50	Counter-Exploitation: RL agents bullied the Tester, refusing to return to cooperation.
Resentful	80.0%	2.61	18.4%	1.14	Critical Divergence: Q-Learning learned to cooperate; SARSA learned to defect.
Self-Play	96.5%	2.94	11.6%	1.35	Emergence of Norms: Q-Learning established trust; SARSA fell into distrust.

- [6] N. Brown and T. Sandholm, "Superhuman AI for multiplayer poker," **Science**, vol. 365, no. 6456, pp. 885–890, 2019.
- [7] T. W. Sandholm and R. H. Crites, "Multiagent reinforcement learning in the iterated prisoner's dilemma," *Biosystems*, vol. 37, no. 1–2, pp. 147–166, 1996.
- [8] L. Xue, C. Sun, D. Wunsch, Y. Zhou, and F. Yu, "An adaptive strategy via reinforcement learning for the prisoner's dilemma game," *IEEE/CAA J. Autom. Sinica*, vol. 5, no. 1, pp. 301–310, Jan. 2018.
- [9] V. Vassiliades and C. Christodoulou, "Multiagent reinforcement learning in the iterated prisoner's dilemma: Fast cooperation through evolved payoffs," in *Proc. Int. Joint Conf. Neural Networks (IJCNN)*, Barcelona, Spain, 2010, pp. 1–8.
- [10] N. Anastassacos and M. Musolesi, "Learning through probing: A decentralized reinforcement learning architecture for social dilemmas," *CoRR*, vol. abs/1809.10007, 2018.
- [11] G. Feehan and S. S. Fatima, "Augmenting reinforcement learning to enhance cooperation in the iterated prisoner's dilemma," in *Proc. 14th Int. Conf. Agents and Artificial Intelligence (ICAART)*, 2022, pp. 146–157.
- [12] M. I. G. Mali, "Finding cooperation in the N -player iterated prisoner's dilemma with deep reinforcement learning over dynamic complex networks," *Procedia Comput. Sci.*, vol. 207, pp. 465–474, 2022.
- [13] P. Barnett and J. Burden, "Oases of cooperation: An empirical evaluation of reinforcement learning in the iterated prisoner's dilemma," in *Proc. SafeAI@AAAI Workshop*, CEUR Workshop Proc., vol. 3087, 2022.
- [14] X. Sun, F. R. Pieroth, K. Schmid, M. Wirsing, and L. Belzner, "On learning stable cooperation in the iterated prisoner's dilemma with paid incentives," in *Proc. 42nd IEEE Int. Conf. Distributed Computing Systems Workshops (ICDCSW)*, 2022, pp. 113–118.
- [15] M. Harper, V. Knight, M. Jones, G. Koutsouvolos, N. E. Glynatsi, and O. Campbell, "Reinforcement learning produces dominant strategies for the iterated prisoner's dilemma," *PLOS ONE*, vol. 12, no. 12, Art. no. e0188046, Dec. 2017.
- [16] Q. Bertrand, J. A. Duque, E. Calvano, and G. Gidel, "Self-play Q -learners can provably collude in the iterated prisoner's dilemma," in *Proc. 42nd Int. Conf. Machine Learning (ICML)*, Proc. Mach. Learn. Res., vol. 267, pp. 3952–3975, 2025.
- [17] C. J. C. H. Watkins and P. Dayan, "Q-learning," *Machine Learning*, vol. 8, no. 3–4, pp. 279–292, 1992.
- [18] Y. Kuroe and H. Iima, "Classes of dilemma problems and their multi-agent reinforcement learning method," *IEEE Access*, vol. 12, pp. 1–20, 2024, doi: 10.1109/ACCESS.2024.3438937.
- [19] M. Cieřla, "Reputation in the iterated prisoner's dilemma: A simple, analytically solvable agents' model," *Entropy*, vol. 27, no. 6, Art. no. 639, 2025.
- [20] R. Willis, Y. Du, J. Z. Leibo, and M. Luck, "Will systems of LLM agents cooperate: An investigation into a social dilemma," *arXiv preprint*, arXiv:2501.16173, 2025.
- [21] Z. Li, "GeoShapley: A Game Theory Approach to Measuring Spatial Effects in Machine Learning Models," *Annals of the American Association of Geographers*, vol. 114, no. 7, pp. 1365–1385, 2024, doi: 10.1080/24694452.2024.2350982.
- [22] Y. Gebreyesus, D. Dalton, S. Nixon, D. De Chiara, and M. Chinnici, "Machine Learning for Data Center Optimizations: Feature Selection Using Shapley Additive exPlanation (SHAP)," *Future Internet*, vol. 15, no. 3, 2023, doi: 10.3390/fi15030088.
- [23] M. Zaki, Jayadeva, and N. M. A. Krishnan, "Extracting processing and testing parameters from materials science literature for improved property prediction of glasses," *Chemical Engineering and Processing – Process Intensification*, vol. 180, 2022, doi: 10.1016/j.cep.2021.108607.
- [24] F. Zhang, R. Deng, X. Zhao, and M. M. Wang, "Load Balancing for Distributed Intelligent Edge Computing: A State-Based Game Approach," *IEEE Transactions on Cognitive Communications and Networking*, vol. 7, no. 4, pp. 1066–1077, 2021, doi: 10.1109/TCCN.2021.3087178.
- [25] Z. Wang, Z. Song, C. Shen, and S. Hu, "Emergence of Punishment in Social Dilemma with Environmental Feedback," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 37, pp. 11708–11716, 2023, doi: 10.1609/aaai.v37i10.2636126383.
- [26] J. Perolat et al., "Mastering the game of Stratego with model-free multiagent reinforcement learning," *Science*, vol. 378, no. 6623, pp. 990–996, 2022, doi: 10.1126/science.add4679.

- [27] C. Fan, C. Zhang, A. Yahja, and A. Mostafavi, "Disaster City Digital Twin: A vision for integrating artificial and human intelligence for disaster management," *International Journal of Information Management*, vol. 56, 2021, doi: 10.1016/j.ijinfomgt.2019.102049.
- [28] M. Zaki, Jayadeva, and A. Krishnan, "Extracting processing and testing parameters from materials science literature for improved property prediction of glasses," *Chemical Engineering and Processing*, vol. 180, 2022, doi: 10.1016/j.cep.2021.108607.
- [29] Axelrod, R. (1984). *The Evolution of Cooperation*. Basic Books.
- [30] Friedman, J. W. (1971). A non-cooperative equilibrium for supergames. *The Review of Economic Studies*, 38(1):1–12.
- [31] Fudenberg, D. and Maskin, E. (1986). The folk theorem in repeated games with discounting or with incomplete information. *Econometrica*, pages 533–554.
- [32] Nowak, M. and Sigmund, K. (1993). A strategy of win-stay, lose-shift that outperforms tit-for-tat in the prisoner's dilemma game. *Nature*, 364(6432):56–58.
- [33] Rummery, G. A. and Niranjan, M. (1994). On-line q-learning using connectionist systems. *University of Cambridge, Department of Engineering Technical Report*.
- [34] Shoham, Y. and Leyton-Brown, K. (2008). *Multiagent systems: Algorithmic, game-theoretic, and logical foundations*. Cambridge University Press.
- [35] Sutton, R. S. and Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press.
- [36] Watkins, C. J. and Dayan, P. (1992). Q-learning. *Machine learning*, 8(3):279–292.